20261112 Quiz

for (MATH261) Discrete Mathematics

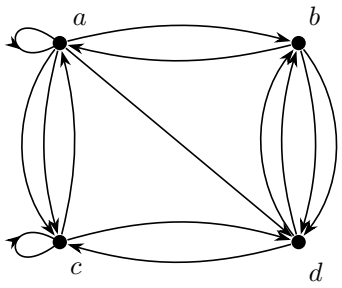
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Problem 1.

(a) Draw the directed graph represented by the given adjacency matrix.

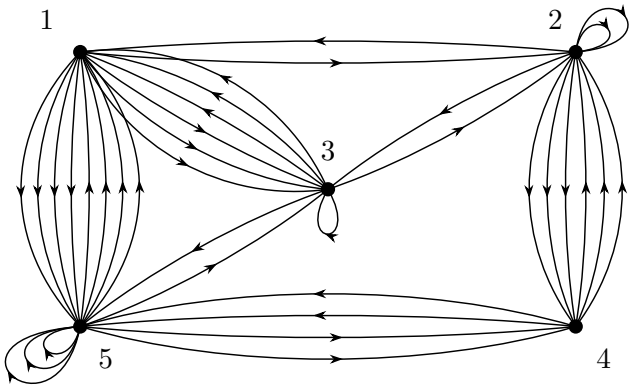
(b) Find the adjacency matrix of the given directed multigraph.

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{pmatrix}.$$



Solution.

(a) The directed multigraph is shown below. Each edge, including each loop, is drawn separately.



(b) With rows and columns ordered as a, b, c, d, the adjacency matrix is

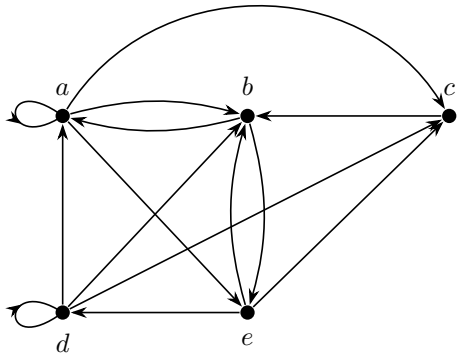
$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \end{pmatrix}.$$

□

Problem 2. Consider the following directed graph.

(a) Find all paths starting at vertex a and ending at vertex b or c. Determine the length of each path.

(b) Find all cycles starting at vertex a. Determine the length of each cycle.



Solution.

(a) A path cannot repeat a vertex. The paths are listed below, grouped by their terminal vertices; the number in parentheses is the length.

Ending at b	Ending at c
(a, b) (1)	(a, c) (1)
(a, c, b) (2)	(a, e, c) (2)
(a, e, b) (2)	(a, b, e, c) (3)
(a, e, c, b) (3)	(a, e, d, c) (3)
(a, e, d, b) (3)	(a, b, e, d, c) (4)
(a, e, d, c, b) (4)	

These 11 paths are exhaustive: the first edge from a goes to b, c, or e, and no vertex may be revisited.

(b) A cycle returns to its initial vertex without repeating any other vertex. The cycles starting at a, grouped by length, are

Length	Cycle
1	(a, a)
2	(a, b, a)
3	(a, c, b, a), (a, e, b, a), (a, e, d, a)
4	(a, b, e, d, a), (a, e, c, b, a), (a, e, d, b, a)
5	(a, c, b, e, d, a), (a, e, d, c, b, a)

There are 10 cycles in total, including the loop (a, a) and the directed 2-cycle (a, b, a) under the course definition. Checking the first edge from a and forbidding repeated intermediate vertices proves exhaustiveness.

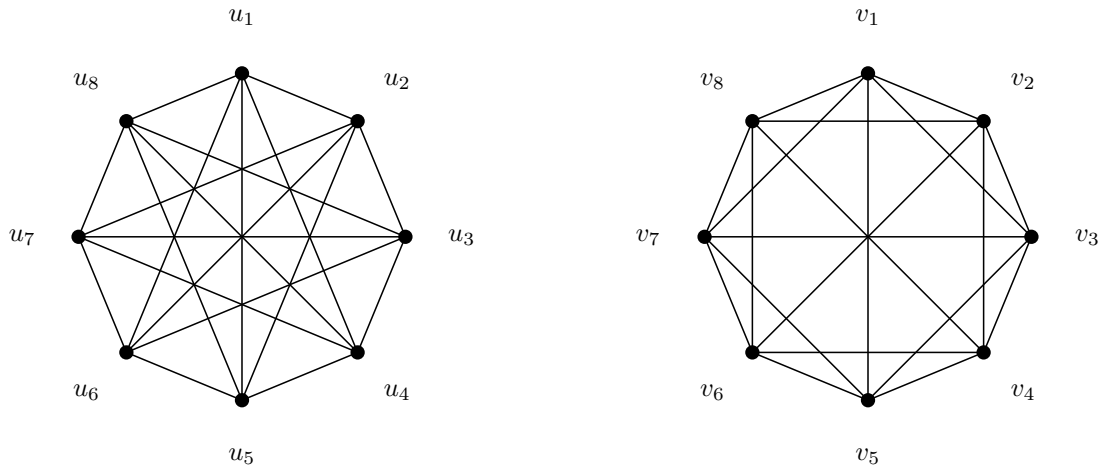
□

Problem 3. Show that isomorphism of simple graphs is an equivalence relation.

Solution. For every simple graph G, the identity on $V(G)$ is an isomorphism, so the relation is reflexive. If $f\colon V(G) \to V(H)$ is an isomorphism, then $f^{-1}\colon V(H) \to V(G)$ is also one, proving symmetry. Finally, if $f\colon V(G) \to V(H)$ and $g\colon V(H) \to V(K)$ are isomorphisms, so is $g \circ f$, proving transitivity. Thus graph isomorphism is an equivalence relation.

□

Problem 4. Determine whether the given pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.



Solution. The graphs are *not* isomorphic. An isomorphism preserves both adjacency and nonadjacency, so it also induces an isomorphism between the complement graphs. Read the vertex subscripts modulo 8. In the left graph, u_i is adjacent to $u_{i\pm1}$, $u_{i\pm3}$, and u_{i+4}. Hence its complement consists of

$$u_1u_3u_5u_7u_1 \qquad \text{and} \qquad u_2u_4u_6u_8u_2,$$

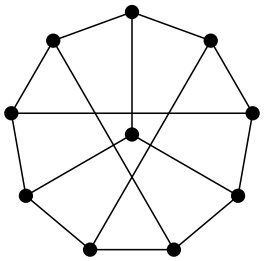
two disjoint 4-cycles. In the right graph, v_i is adjacent to $v_{i\pm1}$, $v_{i\pm2}$, and v_{i+4}, so its complement contains precisely the edges $v_iv_{i\pm3}$. Since $\gcd(3,8) = 1$, these form a single 8-cycle. The complements have different numbers of connected components and therefore cannot be isomorphic. It follows that the original graphs are not isomorphic.

□

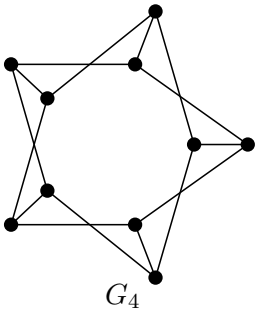
Problem 5. Determine which pairs of the following graphs are isomorphic. For each pair, either exhibit an isomorphism or provide a rigorous argument that none exists.



G_1 G_2



G_3



G_4

Solution. Only $G_2 \cong G_3$.

Step 1. We construct an isomorphism from G_2 to G_3. Label the outer vertices of G_2 by $x_0, \ldots, x_4$, clockwise from the top, and the corresponding inner vertices by $y_0, \ldots, y_4$. Similarly, label the nine outer vertices of G_3 by $z_0, \ldots, z_8$, clockwise from the top, and its central vertex by z_9. Define a bijection f by

Vertex	x_0	x_1	x_2	x_3	x_4
Image	z_0	z_1	z_2	z_3	z_9
Vertex	y_0	y_1	y_2	y_3	y_4
Image	z_8	z_5	z_7	z_4	z_6

The outer 5-cycle of G_2 maps to $z_0z_1z_2z_3z_9z_0$. The five edges joining corresponding outer and inner vertices, called *spokes*, map to z_0z_8, z_1z_5, z_2z_7, z_3z_4, and z_9z_6. The inner 5-cycle (drawn as a star) maps to the edges z_8z_7, z_8z_4, z_5z_4, z_5z_6, and z_7z_6. These are precisely all 15 edges of G_3. Thus f is an isomorphism, and $G_2 \cong G_3$.

Step 2. We show that G_1 is not isomorphic to G_2 or G_3. In G_1, any two adjacent outer vertices, their corresponding inner vertices, and the two spokes form a 4-cycle. By contrast, G_2 has no 4-cycle. Each of its two 5-cycles has no 4-cycle, so a 4-cycle would have to use two spokes and one edge from each 5-cycle. But two consecutive outer indices are never adjacent in the inner pentagram, where vertices are joined when their indices differ by two modulo 5. This is impossible. Since isomorphisms preserve cycle lengths and $G_2 \cong G_3$, neither is isomorphic to G_1.

Step 3. We show that G_4 is not isomorphic to any of the other three graphs. Its edges can be partitioned into those of a 10-cycle and five additional edges joining vertices five steps apart on that cycle. Divide the vertices into two parts by alternating along the 10-cycle. The edges of the cycle join different parts, as do the five additional edges because 5 is odd. Thus G_4 is bipartite. In contrast, G_1, G_2, and G_3 all contain a 5-cycle and hence are not bipartite. Thus G_4 is not isomorphic to any of them.

□